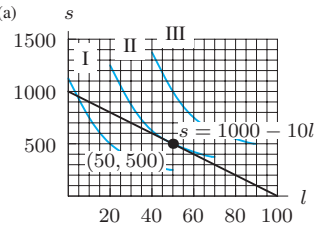
23 (a)



(b) $s = 1000 - 10l$

25 (a) Min; max at endpt of constraint
λ neg
(b) Max; min at endpt of constraint
λ pos

27 $\Delta c/4$; $-\Delta c/4$

31 (a) $C = \$4349$
(b) \$182

33 (a) $W = 225$
$K = 37.5$
(c) $W = 225$
$K = 37.5$
$\lambda = 0.29$

35 (a) No
(b) Yes
(c) $a + b = 1$

37 $x_1 = ((v_1)^{1/2} + (v_2)^{1/2})/(m(v_1)^{1/2})$
$x_2 = ((v_1)^{1/2} + (v_2)^{1/2})/(m(v_2)^{1/2})$

39 (a) $f_1 = \frac{k_1}{k_1+k_2}mg$, $f_2 = \frac{k_2}{k_1+k_2}mg$
(b) Distance the mass stretches the top spring and compresses the lower spring

43 (a) Cost of producing quantity u when prices are p, q
(b) $2\sqrt{pqu}$

45 (a) $-5\lambda^2 + 15\lambda$
(b) 1.5, 11.25
(c) 11.25, 1.5
(d) same

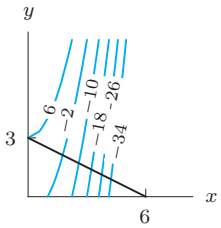
47 (a) $S = \ln(a^a(1-a)^{(1-a)}) + \ln b - a\ln p_1 - (1-a)\ln p_2$
(b) $b = e^c p_1^a p_2^{(1-a)}/(a^a(1-a)^{(1-a)})$

49 Maximum value is 1

51 $f(x, y) = 3x + 4y$

53 $f(x, y) = x^2 + y^2$

55



59 False

61 False

63 False

65 False

67 True

69 True

Chapter 15 Review

1 $(3, -1)$, Saddle point

3 Saddle pt: $(0, -5)$
Local min: $(2, -5)$

5 $(2, 1)$; local min

7 $(2, 3)$: Local and global max

9 Local minimum: $(1, 2)$
Local maximum: $(-1, -2)$
Saddle points: $(1, -2)$ and $(-1, 2)$
No global maximum or minimum

11 Minimum $f(-3/\sqrt{5}, 4/\sqrt{5}) = -5\sqrt{5}$,
Maximum $f(3/\sqrt{5}, -4/\sqrt{5}) = 5\sqrt{5}$

13 Min $= 11.25$; no max

15 Minimum $f(27.907, 23.256) = 1860.484$;
No maximum

17 Maximum $= f(48, 52) = 37{,}600$
No minimum

19 Max $= 21$, no min

21 No global extrema.

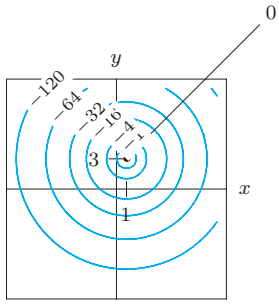
23 Maximum

25 Both

27 0.204

29 $\Delta c/2$; $\Delta c/(2\sqrt{2})$

31

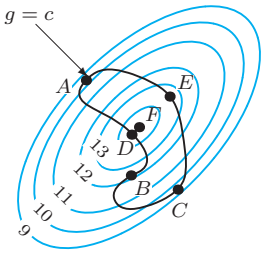


33 (a) (i) Power function
(ii) Linear function
(b) $\ln N = 1.20 + 0.32\ln A$
Agrees with biological rule

35 (a) $2c^2/3$
(b) $4c/3$
(c) $\lambda = m'(c)$

37 $c\left(\frac{aK}{(a+b)P_1}\right)^a\left(\frac{bK}{(a+b)P_2}\right)^b$

39 (a) Points A, B, C, D, E
(b) Point F
(c) Point D



41 $f(x, y, z) = \frac{|Aa+Bb+Cc+D|}{\sqrt{A^2+B^2+C^2}}$

43 $K = 20$
$L = 30$
$C = \$7{,}000$

47 Base: 4 cm by 4 cm
Height: 2 cm

49 $-1/2$

51 (a) $x = 1/4 - a$, $y = 1$, saddle point

53 (a) 21.0208
(b) $g = 20.5, 21.2211$; $g = 20.2, 21.1008$
(c) $g = 20.5, 21.2198$; $g = 20.2, 21.1007$

Section 16.1

1 24; 43.5

3 Over: Approx 137
Under: Approx 60

5 about 2300

7 Positive

9 Zero

11 Zero

13 Positive

15 25.2°C

17 Need f nonnegative everywhere

19 $f(x, y) = 5 - x - y$; R is square with vertices $(\pm 1, \pm 1)$

21 False

23 False

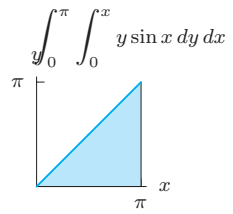
25 True

27 True

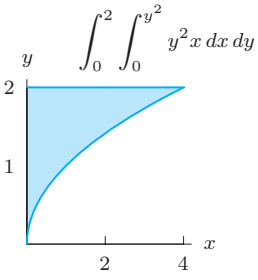
29 False

Section 16.2

1



3



5 150

7 54

9 $e - 2$

11 $3 - \sin 3$

13 $(e^4 - 1)(e^2 - 1)e$

15 -2.678

17 $\int_1^4 \int_1^2 f\,dy\,dx$ or $\int_1^2 \int_1^4 f\,dx\,dy$

19 $\int_{-1}^3 \int_{-2}^{(1-3x)/4} f\,dy\,dx$
or $\int_{-2}^1 \int_{-1}^{(1-4y)/3} f\,dx\,dy$

21 $\int_1^3 \int_{\frac{1}{2}(y-1)}^{-\frac{1}{2}(y-5)} f\,dx\,dy$

23 $\frac{4}{15}(9\sqrt{3}-4\sqrt{2}-1) = 2.38176$

25 32/9

27 13/6

29 0

31 2/3

33 $(e-1)/2$

35 $\frac{2}{9}(3\sqrt{3}-2\sqrt{2})$

37 $\frac{1}{2}(e^2-1)$

39 (a) 8/3
(b) 16/3

41 (a) $\int_0^{1/2}\int_y^{1-y} f(x,y)\,dx\,dy$
$\int_0^{1/2}\int_0^{x} f(x,y)\,dy\,dx$
$\int_{1/2}^{1}\int_0^{1-x} f(x,y)\,dy\,dx$
(b) 1/8

43 15

45 $\int_{-5}^{5}\int_{-\sqrt{25-y^2}}^{\sqrt{25-y^2}}(25-x^2-y^2)\,dx\,dy$

47 $\int_0^4\int_{y-4}^{(4-y)/2}(4-2x-y)\,dx\,dy$

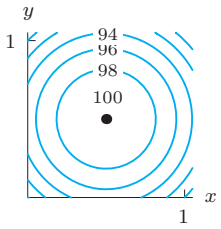
49 1/3

51 16/3

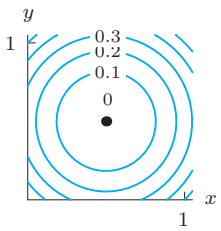
53 Volume $= 1/(6abc)$

55 (a) Circles centered at $(1,0)$
(b) $\int_{-\sqrt{3}}^{\sqrt{3}} e^{-y^2}\,dy$
(c) $\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} e^{-(x-1)^2-y^2}\,dy\,dx$

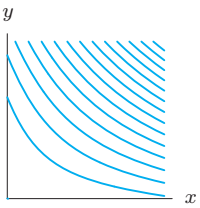
57 (a)



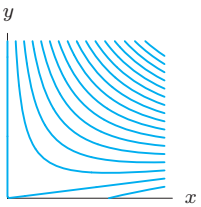
(b)



59 (a) $(4/3)a + b + (4/3)c = 20$
(b) $f(x,y) = x^2 + \frac{44}{3}xy + 3y^2$:



$f(x,y) = -3x^2 + 24xy$:



61 $k(a^3b + ab^3)/3$

63 Integrals not over same region

65 $\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} 2\,dx\,dy$

\+ 67 $\int_0^2\int_0^{6-3y} 1\,dx\,dy$

69 False

71 False

73 False

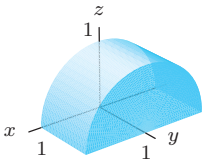
75 False

Section 16.3

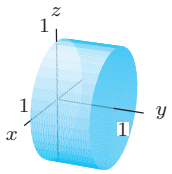
1 2

3 −8

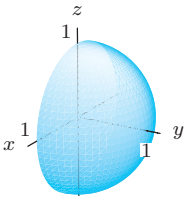
5



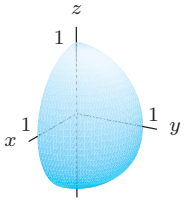
7



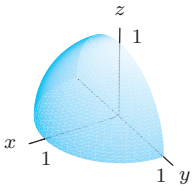
9



11



13



15 Positive

17 Positive

19 Zero

21 Positive

23 Zero

25 Positive

27 Positive

29 125

31 (a) $2\int_0^{10}\int_0^1\int_y^1 e^{-3x}\,dz\,dy\,dx$
Other answers are possible
(b) $(1-e^{-30})/3$

33 $V = \int_0^1\int_0^2\int_{x+y}^{1+2x+2y} 1\,dz\,dy\,dx$
Can reverse order x, y

35 $V = \int_0^1\int_0^{1-x}\int_{6-3x-4y}^{6-2x-2y} 1\,dz\,dy\,dx$
Can reverse order x, y

37 $V = \int_{-\sqrt{5}}^{\sqrt{5}}\int_{-\sqrt{5-x^2}}^{\sqrt{5-x^2}}\int_2^{\sqrt{9-x^2-y^2}} 1\,dz\,dy\,dx$
Can reverse order x, y

39 $\int_0^1\int_{-r}^{r}\int_0^{\sqrt{r^2-x^2}} f(x,y,z)\,dz\,dx\,dy$

41 $\int_{-r}^{r}\int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}}\int_0^{\sqrt{r^2-x^2-z^2}} f(x,y,z)\,dy\,dz\,dx$

43 1/2

45 162

47 1

49 29 gm

51 243

53 (a) $x+y+z=1$
(b) 1/6

55 $\int_0^2\int_0^{1-\frac{z}{4}}\int_0^{2-2y-\frac{z}{2}} f(x,y,z)\,dx\,dy\,dz$

57 $\int_{-4}^{4}\int_{-\sqrt{16-x^2}}^{\sqrt{16-x^2}}\int_3^{\sqrt{25-x^2-y^2}} dz\,dy\,dx$

59 $\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_{-\sqrt{1-z^2}}^{\sqrt{1-z^2}} dy\,dz\,dx$

61 $\int_0^2\int_0^{6y^2}\int_0^{\sqrt{12-3y^2}} f(x,y,z)\,dx\,dz\,dy$

63 4

65 $m = 2$;
$(\bar{x},\bar{y},\bar{z}) = (13/24, 13/24, 25/24)$

67 $5m/3$

71 Need same limits on innermost integral

73 $f(x,y,z) = z$

75 True

77 True

79 False

81 True

83 True

Section 16.4

1 $\int_0^{\pi/2}\int_0^{1/2} f\,r\,dr\,d\theta$

3 $\int_{\pi/4}^{3\pi/4}\int_0^2 f\,r\,dr\,d\theta$